

NON SCAN:

VERILOG:

```
module dff_8bit (  
    input clk,  
    input rstn,          // Active LOW reset (aligned with your counter style)  
    input [7:0] d,  
    output reg [7:0] q  
);  
always @(posedge clk) begin  
    if (!rstn)  
        q <= 8'b00000000;  
    else  
        q <= d;  
end  
endmodule
```

TEST BENCH:

```
module tb_dff_8bit;  
    reg clk;  
    reg rstn;  
    reg [7:0] d;  
    wire [7:0] q;  
    dff_8bit uut (  
        .clk(clk),  
        .rstn(rstn),  
        .d(d),  
        .q(q)  
    );  
    // Clock generation (100 MHz)  
    always #5 clk = ~clk;  
    initial begin  
        $dumpfile("dff8_activity.vcd");  
        $dumpvars(0, tb_dff_8bit);  
        // Initialize  
        clk = 0;  
        rstn = 0;  
        d = 8'b00000000;  
        // Release reset  
        #20 rstn = 1;  
        // Stimulus  
        #10 d = 8'b10101010;  
        #10 d = 8'b11110000;  
        #10 d = 8'b00001111;  
        #10 d = 8'b11001100;  
        // Toggle reset again (to increase switching activity)  
        #20 rstn = 0;  
        #10 rstn = 1;  
    end  
endmodule
```

```
#30 $finish;  
end  
endmodule
```